

INTRODUCTION TO STATISTICAL METHODS

Correlation vs Time Series

Cross-sectional vs time-series data, components, autocorrelation, and spurious correlation.

Companion reader · Read beside the slides

How to use this companion. Colour boxes guide you: **blue** intuition, **amber** everyday picture, **green** worked example, **red** watch out, **purple** key takeaway. Read it next to the slides, in order.

1 Cross-sectional vs time-series data

Ordinary correlation compares two variables measured across many *subjects* at one time (**cross-sectional**). A **time series** is one variable measured on the *same* subject over *time* — and time order matters. The key new idea is that a series can be correlated with its *own past*: **autocorrelation**.

The intuition

Today's temperature is a good clue to tomorrow's — consecutive values are related. That self-similarity over time is exactly what cross-sectional correlation ignores and time-series analysis exploits.

2 The components of a time series

A series is usually seen as a combination of four parts: a long-run **trend** (steady rise/fall), a repeating **seasonal** pattern (fixed period, e.g. monthly), a longer **cyclical** swing, and **irregular** (random) noise. Separating these is the heart of forecasting.

3 Autocorrelation

The **autocorrelation** at lag k is the correlation between the series and a copy of itself shifted by k steps:

$$r_k = \frac{\sum_{t=1}^{n-k} (x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2}.$$

High r_1 means consecutive values move together (**serial correlation**).

Worked example: lag-1 autocorrelation

Series $x = (2, 4, 6, 8, 10)$, mean $\bar{x} = 6$. Numerator = $\sum (x_t - 6)(x_{t+1} - 6) = (-4)(-2) + (-2)(0) + (0)(2) + (2)(4) = 8 + 0 + 0 + 8 = 16$. Denominator = $\sum (x_t - 6)^2 = 16 + 4 + 0 + 4 + 16 = 40$. So $r_1 = \frac{16}{40} = 0.4$ — a positive lag-1 autocorrelation, as expected for a steadily rising series.



Figure 1: A time series often decomposes into a trend plus a repeating seasonal pattern (plus noise) — unlike cross-sectional data, the horizontal axis is time.

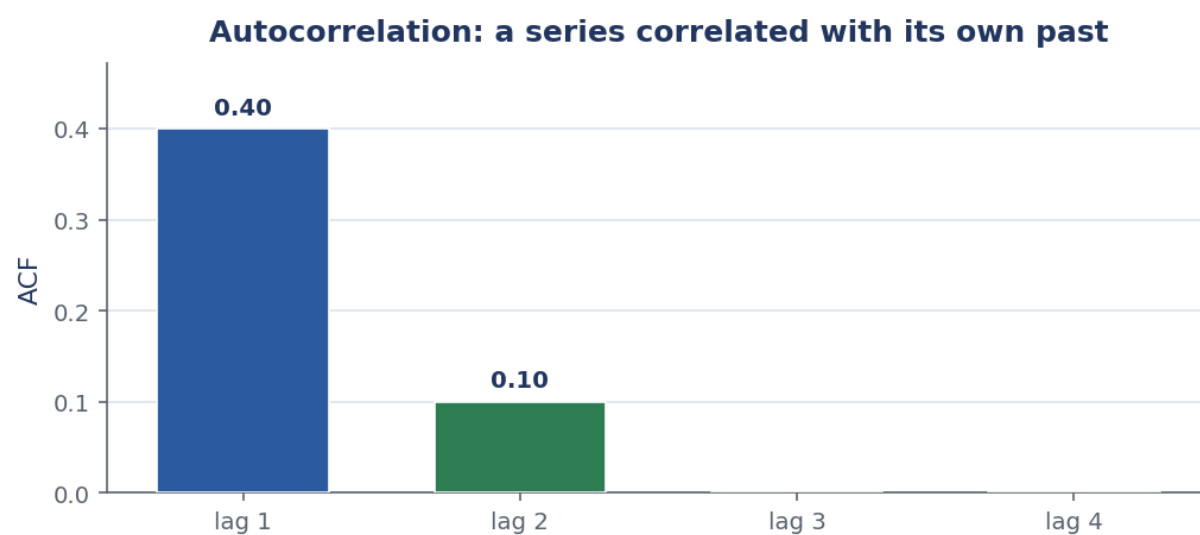


Figure 2: Plotting autocorrelation against lag (the **correlogram**) reveals structure: slowly-decaying values signal a trend; a spike at the seasonal lag signals seasonality.

✗ Watch out

Spurious correlation: two unrelated series that both trend upward over time (ice-cream sales and drownings) can show a high correlation with no causal link. When correlating time series, remove trends first (e.g. by differencing) before trusting the number.

✓ Key takeaway

Cross-sectional correlation compares variables across subjects; a **time series** tracks one variable over time and can correlate with its *own past* (**autocorrelation** r_k). A series decomposes into **trend**, **seasonality**, **cycles** and **noise**. Worked lag-1 for (2, 4, 6, 8, 10) is $r_1 = 0.4$. Beware **spurious correlation** between two trending series — detrend first.